

TRADUZIONE PROGRAMMI RISC-V

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TRADUZIONE PROVA SCRITTA 14 GIUGNO 2022 - SOMMA VETTORI

MAIN:

ADDI x2, x2, -16

111111110000 00010 000 00010 0010011
0xFF01 0113

SD x1, 0(x2)

00000000 00001 00010 011 00000 0100011
0x0011 3023

SD x8, 8(x2)

imm = 0000 0000 1000

00000000 01000 00010 011 01000 0100011
0x0081 3423

LA x10, y

CODIFICA NON DISPONIBILE

LA x11, x

CODIFICA NON DISPONIBILE

ADDI x12, x0, 4

000000000100 00000 000 01100 0010011
0x0040 0613

ADDI x13, x0, 0

000000000000 00000 000 01101 0010011
0x0000 0603

SAL x1, MYSTRCPY → ≠ AVANTI → ≠ 4 = 28 (in 20 bit)

0000 0000 0000 0001 1100 >> 1

0000 0000 0000 0000 1110

000000011100 00000000 00001 1101111
0x0100 00EF

LD x1, 0(x2)

000000000000 00010 011 00001 0000011
0x0001 3083

LD x8, 8(x2)

0000000001000 00010 011 01000 0000011
0x0081 3403

ADDI x2, x2, 16

0000|0001|0000|0001|0000|0001|0010|0011

0x0101 0113

LI x10, 0 → ADDI x10, x0, 0

0000|0000|0000|0000|0000|0000|0101|0011

0x0000 0513

LI x17, 93 → ADDI x17, x0, 93 (64+16+8+4+1)

0000|0101|1101|0000|0000|1000|1001|0011

0x0500 0893

ECALL CODIFICA NON DISPONIBILE

MYSTRCPY:

ADD x29, x0, x0

0000|0000|0000|0000|0000|1110|0110|0011

0x0000 0EB3

BNE x13, x0, LEN → 8 AVANTI → $8 \cdot 4 = 32$ (1112 BIT)

0000 0010 0000 >> 1

0000 0001 0000

0000|0001|0000|0110|0010|0000|1100|0011

0x0206 9063

DIR:

BGE x29, x12, END 25 AVANTI $25 \cdot 4 = 100$ (12 BIT)

$100 = 64 + 32 + 4$

0000 0110 0100 >> 1

0000 0011 0010

0000|0110|0110|1110|1010|0010|1100|0011

0x06CE D263

ADD x5, x29, x11

0000|0000|0101|1110|0000|0010|0110|0011

0x00BE 82B3

LB x5, 0(x5)

0000|0000|0000|0000|0010|0000|0010|0011

0x0002 8283

BEQ x5, x0, END → 22 AVANTI → 22 · 4 = 88 (in 12 bit)

$$88 = 64 + 16 + 8$$

0000 0101 1000 >> 1

0000 0010 1100

0000010 00000 00101 000 11000 1100011

0x0402 8C63

ADD x6, x20, x10

0000000 01010 11101 000 00110 0110011

0x00AE 8333

SB x5, 0(x6)

0000000 00101 00110 000 00000 0100011

0x0053 0023

ADDI x20, x20, 1

000000000001 11101 000 11101 0010011

0x001E 8E93

SAL x0, DIR → 7 INDIETRO → -7 · 4 = -28 (in 20 bit)

(28)₁₀ = 0000 0000 0000 0001 1100 ~

1111 1111 1111 1110 0011 + 1

(-28)₁₀ = 1111 1111 1111 1110 0100 >> 1

1111 1111 1111 1111 0010

11111110010111111111 00000 1101111

0xF5EF F06F

LEN:

ADD x5, x20, x11

0000000 01011 11101 000 00101 0110011

0x00BE 82B3

LB x5, 0(x5)

000000000000 00101 000 00101 0000011

0x0002 8283

BEQ x5, x0, HREV

3 AVANTI

3 · 4 = 12 (12 bit)

0000 0000 1100 >> 1

0000 0000 0110

0000000 00000 00101 000 01100 1100011

0x0002 8663

ADDI x29, x29, 1

0000 0000 0001 1110 1000 1110 0010 0011

0x001E 8E93

SAL x0, LEN 4 INDIETRO -4 = -16 (16 20 BIT)

1111 1111 1111 1111 0000 >> 1

1111 1111 1111 1111 1000

1111 1111 0001 1111 1111 0000 1101 1111

0xFF1E F06F

HREV:

ADD x30, x0, x0

0000 0000 0000 0000 0000 000 1111 0110 0011

0x0000 0F33

BGE x12, x29, REV → 2 AVANTI → 2 · 4 = 8 (16 12 BIT)

0000 0000 1000 >> 1

0000 0000 0100

0000 000 1110 1011 0110 101 0100 1100 0011

0xC1D6 5463

ADD x29, x0, x12

0000 000 0110 0000 000 1110 1011 0110 0011

0x00C0 0EB3

REV:

BGE x30, x29, END → 9 AVANTI → 9 · 4 = 36 (12 BIT)

0000 0010 0100 >> 1

0000 0001 0010

0000 001 1110 1111 101 0010 1100 0011

0x03DF 5263

ADDI x5, x29, -1

1111 1111 1111 1110 1000 0010 1001 0010 0011

0xFFFE 8293

SUB x5, x5, x30

0100 000 1111 0010 1000 0010 1011 0110 0011

0x41E2 82B3

ADD x5, x5, x11

00000000 01011 00101 000 00101 0110011

0x00B2 82B3

LB x5, 0(x5)

00000000 0000 00101 000 00101 0000011

0x0002 8283

ADD x6, x30, x10

00000000 01010 11110 000 00110 0110011

0x00AF 0333

SB x5, 0(x6)

00000000 00101 00110 000 00000 0100011

0x0053 0023

ADDI x30, x30, 1

00000000 0001 11110 000 11110 0010011

0x001F 0F13

SAL x0, REV $\rightarrow$ 8 INDIETRO $\rightarrow -8 \cdot 4 = -32$ (in 20 BIT)

1111 1111 1111 1110 0000 $\gg 1$

1111 1111 1111 1111 0000

11111110000111111111 00000 1101111

0xFE1F F06F

END:

ADD x5, x29, x10

00000000 01010 11101 000 00101 0110011

0x00AE 82B3

SB x0, 0(x5)

00000000 00000 00101 000 00000 0100011

0x0002 8023

SALR x0, 0(x1)

000000000000 00001 000 00000 1100111

0x0000 8067

- RICORDA CHE GLI OFFSET NEGATIVI SONO RAPPRESENTATI IN COMPLEMENTO A DUE.
- RICORDA CHE LA CODIFICA DELLE STORE È: $SX \ RS_2, \text{OFF}(RS_1)$